

THE EFFECT OF BATCH SIZE AND MOMENTUM ON THE PER-STEP PERFORMANCE OF THE MUON OPTIMIZER

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ABSTRACT

Muon (Jordan et al., 2024) is a recently proposed optimizer that applies a Newton–Schulz orthogonalization to the momentum-buffered gradient of hidden weight matrices. We study, empirically, how the two hyperparameters that directly shape Muon’s update – the mini-batch size B and the momentum coefficient μ – affect its *per-step* progress on a Wide ResNet 28–10 / CIFAR-10 benchmark. Sweeping a 6×4 grid ($B \in \{64, \dots, 2048\}$, $\mu \in \{0.8, 0.9, 0.95, 0.99\}$), we find that (i) Muon scales monotonically with B in steps – well past the bs=256 saturation point typical for vanilla SGD – but (ii) at the cost of sample efficiency and ~ 2 pp of final test accuracy. The momentum coefficient is roughly insensitive at small B but becomes the dominant hyperparameter at large B , where the EMA horizon must be matched to the (now short) training budget. We give a simple sample-horizon heuristic, $\tau \approx B/(1 - \mu)$, that captures the observed best configurations.

1 INTRODUCTION

Most popular optimizers for deep learning – SGD with momentum, Adam (Kingma & Ba, 2015), AdamW (Loshchilov & Hutter, 2019) – update each parameter *independently*. Muon (Jordan et al., 2024), by contrast, replaces the gradient (or its momentum buffer) of every hidden weight matrix by its closest orthogonal matrix, computed efficiently by a quintic Newton–Schulz iteration. This makes Muon a practical, matrix-aware second-order-flavoured method, in the same family as Shampoo (Anil et al., 2020) and modular-duality methods (Bernstein & Newhouse, 2024), but with cost comparable to a few extra matmuls per step. Muon recently set speed records for training small GPT models and is increasingly used as a drop-in replacement for AdamW on hidden weights.

Two hyperparameters directly enter the update: the mini-batch size B and the momentum coefficient μ of the EMA fed to the Newton–Schulz step. The optimization literature has a well-developed picture of B for SGD/Adam: increasing B reduces gradient variance and accelerates per-step progress, but only up to a critical batch size beyond which the scalability gain vanishes (McCandlish et al., 2018; Shallue et al., 2019). Whether and how this picture transfers to Muon is a priori unclear: the orthogonalization step strips magnitude information from the update, which is precisely the dimension along which variance reduction operates. We therefore ask: *at fixed compute, how does Muon’s per-step progress depend on B , and how does μ interact with that scaling?*

Contribution. We run a controlled 6×4 sweep of (B, μ) for WRN-28–10 on CIFAR-10 (Section 2) and report (Section 3): (a) per-step convergence becomes monotonically faster with B up to bs=2048, with no clear saturation in steps; (b) per-sample efficiency degrades sharply with B ; (c) the optimal μ moves from a broad plateau around 0.9–0.95 at small B to a sharp $\mu^* \approx 0.95$ at large B , where both under- (0.8) and over-smoothing (0.99) hurt. We give a simple EMA-horizon argument for (c). Code at `plot.py`, `train.py`, `sweep.py`.

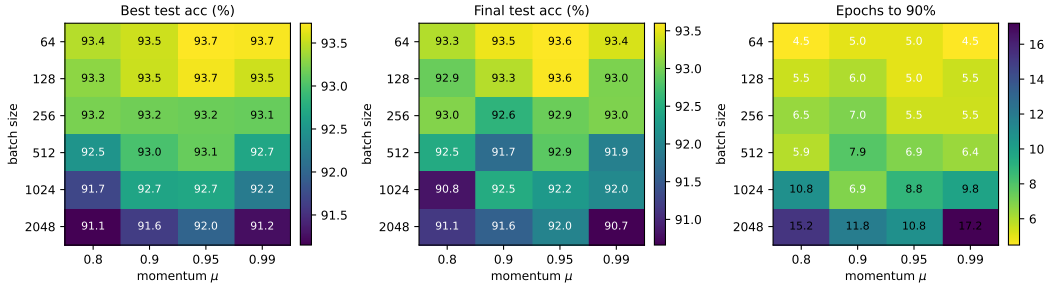


Figure 1: Sweep summary on the (B, μ) grid. *Left*: best test accuracy during the 25-epoch run. *Centre*: final-evaluation accuracy. *Right*: epochs to first cross 90% test accuracy. Best accuracy peaks at small/medium B ; fastest configurations combine large B with $\mu \in [0.9, 0.95]$.

2 BACKGROUND AND METHOD

Dataset. CIFAR-10 (Krizhevsky, 2009): 60,000 32×32 colour images in 10 classes (50,000 train / 10,000 test). Pixels are normalised channel-wise; the training stream uses standard CIFAR augmentation (4-pixel reflection padding, random 32×32 crop, random horizontal flip), the test stream none.

Model. Wide ResNet 28–10 (Zagoruyko & Komodakis, 2016; He et al., 2016): a 3×3 stem conv, three stages of pre-activation residual blocks with widths $\{160, 320, 640\}$ and batch normalisation (Ioffe & Szegedy, 2015), global average pooling, and a 10-way linear classifier (~ 36.5 M parameters, no dropout, channels-last layout). WRN-28–10 is a standard CIFAR-10 backbone; our goal is not absolute accuracy but a fair, controlled comparison across (B, μ) .

Muon. For each 2D weight matrix W with stochastic gradient g_t , Muon maintains an EMA-style buffer $m_t = \mu m_{t-1} + g_t$, forms a Nesterov-style direction $u_t = g_t + \mu m_t$, and updates

$$W_{t+1} = W_t - \eta s \text{NS}_5(u_t), \quad s = \max(1, (d_{\text{out}}/d_{\text{in}})^{1/2}), \quad (1)$$

where NS_5 is the 5-step Newton–Schulz iteration of Jordan et al. (2024) that maps a matrix $X = U\Sigma V^\top$ to a numerical approximation of $UV^\top$. Conv weights $(c_{\text{out}}, c_{\text{in}}, k_h, k_w)$ are reshaped to $(c_{\text{out}}, c_{\text{in}} k_h k_w)$ before NS_5 is applied. Operationally, $\text{NS}_5(u_t)$ is the closest-orthogonal matrix to u_t : Muon *normalises every singular value to one*, so the update is purely a direction. Non-matrix parameters (biases, BN parameters, the stem conv and the final classifier) are trained with AdamW (Loshchilov & Hutter, 2019), as in the reference implementation.

Training pipeline and sweep. We fix the Muon learning rate $\eta=0.02$, AdamW $\eta=3 \times 10^{-4}$ with $\beta=(0.9, 0.95)$, weight decay 5×10^{-4} , and 5 Newton–Schulz steps – all defaults of the reference implementation. Every (B, μ) configuration trains for exactly 25 *epochs* (so all see the same data; smaller batches take more steps), with test accuracy logged twice per epoch. We sweep $B \in \{64, 128, 256, 512, 1024, 2048\}$ and $\mu \in \{0.8, 0.9, 0.95, 0.99\}$ – a 24-cell grid, single seed, ~ 2.7 h on an NVIDIA GTX 1660 Ti. $\mu=0.95$ is the reference value of Jordan et al. (2024).

3 RESULTS

Per-step performance scales monotonically with B . At fixed μ , larger batches reach every accuracy threshold in fewer optimizer steps (Fig. 2, top). Steps to 90% drop from $\sim 3,510$ (bs=64) to ~ 264 (bs=2048) at $\mu = 0.95$ – a $13.3 \times$ reduction for a $32 \times$ batch increase (Fig. 2, bottom-left). A log–log fit gives $\text{steps}_{90} \propto B^{-0.74}$: sub-linear, but *no plateau within our range*. For SGD/Adam on similar CIFAR setups the critical batch size sits around $B^* \sim 256\text{--}512$ (Shallue et al., 2019); Muon pushes that point at least an order of magnitude further. We attribute this to Muon’s update being magnitude-free ($\text{NS}_5(u_t)$ is orthogonal): SGD’s main benefit from larger B – shrinking gradient *magnitude* noise – is moot for Muon, but the residual gain of cleaning up the singular *directions* that NS_5 picks out persists.

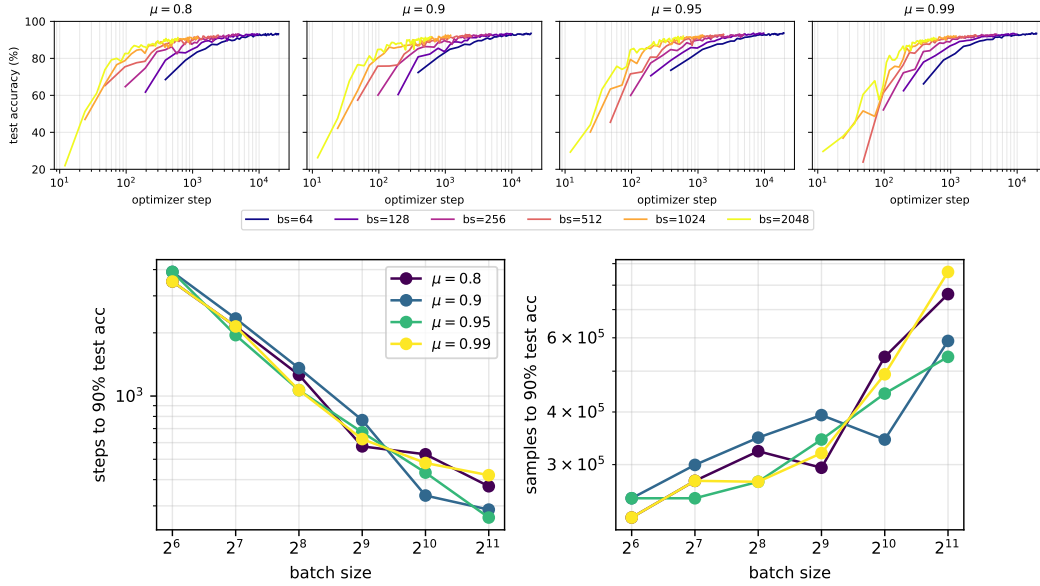


Figure 2: *Top*: test accuracy vs. optimizer step (log), one panel per momentum; at fixed μ larger batches reach every accuracy threshold in fewer steps ($\sim 13\times$ between bs=64 and bs=2048 at 90%). The $\mu=0.99$ panel shows the EMA-horizon pathology – at bs=2048 the $\sim 1/(1-\mu)=100$ -step horizon eats 1/6 of the budget. *Bottom*: cost to first reach 90% test accuracy in optimizer steps (left) and training samples (right), as a function of B , log-log. Steps-to-target falls sub-linearly with B ; samples-to-target *increases*.

Sample efficiency degrades with B , and so does final accuracy. Plotted in samples (Fig. 2, bottom-right) the story flips: 90% takes ~ 250 k samples at bs=128 but ~ 540 k at bs=2048 ($2.2\times$). Mean best test accuracy across the four momenta drops from 93.6% (bs=64) to 91.5% (bs=2048), a 2.1 pp loss (Fig. 1, left). Two effects compound: large batches see fewer noisy iterates, weakening implicit regularisation (Goyal et al., 2017); and at fixed 25 epochs each large-batch update is asked to do strictly more work. Muon does not escape these well-known failure modes (Shallue et al., 2019).

The momentum coefficient becomes critical at large B . At bs=64 the four momenta land within 0.3 pp (93.3–93.7%); at bs=2048 the spread is 0.9 pp (91.1–92.0%), and the curvature in μ flips. At *small* B best accuracy is essentially flat in μ : per-step variance is high, the orthogonalization recovers a clean direction every step, and the buffer adds little. At *large* B the optimum is sharply at $\mu=0.95$ and *both* 0.8 and 0.99 are visibly worse. Under-smoothing ($\mu=0.8$, horizon $1/(1-\mu) \approx 5$ steps) leaves NS_5 acting on a still-noisy buffer; over-smoothing ($\mu=0.99$, ~ 100 -step horizon) outruns the budget (600 steps at bs=2048) so stale gradients dominate. A heuristic falls out: the buffer averages over $\tau \approx B/(1-\mu)$ samples, and the configurations within 0.3 pp of the global best all sit in $\tau \in [10^3, 2 \cdot 10^4]$. This suggests retuning $1-\mu$ as $1/B$; $\mu=0.95$ sits in the middle of that band, which explains why it works across many workloads.

4 DISCUSSION AND CONCLUSION

The two hyperparameters admit a unified picture. Muon’s update is a normalised direction, so variance reduction from a larger B keeps helping past the SGD plateau – there is no magnitude-noise floor – but at the usual large-batch price in samples and regularisation. The buffer has a sample-effective horizon $\tau \approx B/(1-\mu)$; keeping τ constant (not μ) is the clean recipe for retuning momentum once B is chosen, rather than treating 0.95 as universal. Limitations: a single seed, fixed η , fixed-epoch budget, one model/dataset. A joint (B, μ, η) sweep on a transformer – where Muon’s appeal is sharpest – is the most useful follow-up.

STATEMENT OF CONTRIBUTIONS

We hereby state that all the work presented in this report is that of the authors.

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